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Overview of Vulnerability Assessment (VA) Methodology

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Abstract

The evaluation of an existing or proposed Physical Protection System (PPS) requires a methodical approach in which the ability of the security system to meet defined protection objectives is measured. Without this kind of careful assessment, valuable resources might be wasted on unnecessary protection or, worse yet, fail to provide adequate protection of nuclear materials, items, or facilities (targets) against a malicious attack (theft or sabotage) by the defined threat. The Vulnerability Assessment (VA) methodology was developed to implement performance-based physical security concepts at nuclear sites and facilities.

Overview of Vulnerability Assessment (VA) Methodology

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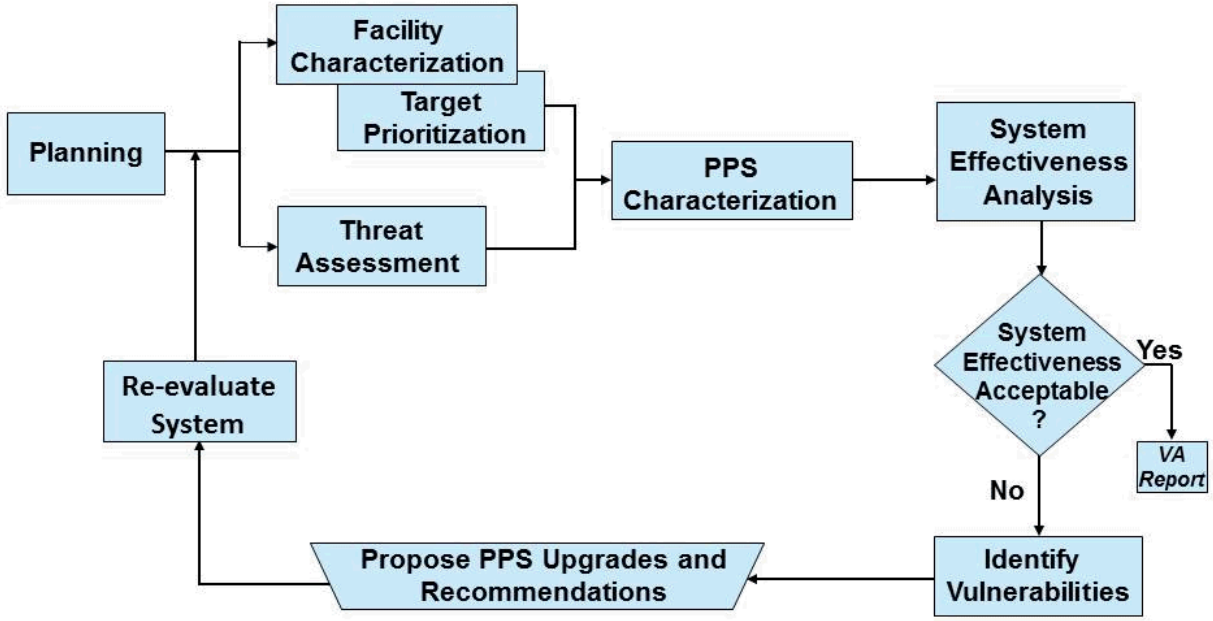


Figure 1. Process Flow Diagram for Vulnerability Assessment

The VA methodology used consists of the following 12 steps:

1. Identify the VA team
2. Plan the VA
3. Define the threat (design basis threat)
4. Characterize the facility
5. Identify the targets and prioritize targets based on appropriate nuclear categorization tables
6. Define the physical protection system
7. Conduct probability of interruption analysis (determine the most vulnerable paths and develop worse-case scenarios)

8. Conduct probability of neutralization analysis
9. Evaluate PPS effectiveness and determine baseline system effectiveness
10. Identify vulnerabilities, areas for improvement, and propose upgrades
11. Re-evaluate the system with proposed improvements (determine upgraded system effectiveness)
12. Report results and recommendations

Step 1. Identify the Vulnerability Assessment Team. An effective VA requires a highly trained team with extensive knowledge and experience in physical protection and assessment. The team requires (1) a project lead that can manage the entire VA process and ensure the assessment and results are correct and (2) specialized experts in the fields of detection, assessment, delay, response forces, analysis, and cyber security.

Step 2. Plan the VA. Once the team is assembled, the project lead develops a project plan that outlines how the VA will be conducted. The project lead identifies team members' roles and responsibilities, develops the VA schedule, and gathers and distributes preliminary data and information to prepare for the facility characterization survey. Several project team meetings are conducted prior to the team's departure to the facility.

Step 3. Define the Threat. Understanding the capabilities and attributes associated with the defined threat is vital to the design and/or upgrade of a PPS. Initially a broad threat spectrum is categorized into a specific subset of adversary unit size, skills, tactics, weapons, and capabilities, and forms the basis for designing the PPS. This subset is known as the Design Basis Threat (DBT).

Step 4. Characterize the Facility. This process involves identifying and understanding the overall facility mission, nuclear operations and processes, company or state legal issues, and the physical conditions of the facility. All information and data collected is used to fully understand what must be protected (targets), from whom (threat), and to what performance level. Interviews are conducted at the facility to understand all aspects of the site and its operations. Physical protection and cyber security specialists conduct thorough site surveys to collect site data on essential PPS and cyber security elements.

Step 5. Identify and Prioritize Targets. Target analysis involves identifying nuclear materials, items, and facilities at the site that must be protected to ensure a high level of system effectiveness against the DBT. These nuclear targets, processes, and operations are identified during the facility characterization phase as potential adversary targets and then prioritized based on appropriate nuclear categorization tables.

Step 6. Define the PPS. A detailed PPS characterization includes identifying facility security elements that provide detection, assessment, delay, and response capabilities. A thorough on-site survey allows security specialists to identify all protection elements and

collect performance data for each component and the system as a whole. Performance-based data for identified security elements are incorporated into a computer model of the facility. If performance-based data cannot be collected at the site, then database values from accepted and validated field tests and subject matter expert judgments are used to generate measures for the security elements.

Step 7. Conduct Probability of Interruption Analysis. The site data collected and assumptions agreed to in Steps 4 and 6 (Facility and PPS Characterization) are used to construct models of all credible pathways from outside the facility perimeter to the targets of concern. A computer model is used to construct an Adversary Sequence Diagram (ASD), which is a functional representation of the PPS at the facility and describes specific protection elements at the facility. Once the ASD is created and populated with site data, the model is used to mathematically calculate the probabilities of interruption for each path (accumulated probabilities of assessed detection and delay times along the pathway), and the results are used to determine the most vulnerable adversary paths. The software default database values are from actual performance tests conducted to determine sensor probabilities of detection and barrier delay times as functions of adversary defeat techniques and tactics. These paths are subsequently reviewed to confirm credibility of detection probabilities and delay times. Any modification made to default values are based on site-specific performance tested data and/or subject matter expert engineering judgment. The justifications for modifications to default values are documented.

Worst-case scenarios are developed by combining the most vulnerable paths with the facility's planned system response to an attack by the DBT. Detection probabilities and path delays are reviewed and modified, if necessary, to validate credibility and reflect real-world conditions. All potential and credible attack options are studied to ensure that worse-case attacks and other serious options that need to be protected against are identified. For the analysis, attack scenarios are described step-by-step, including when the adversary is detected, all adversary task times, and when the response force assesses and communicates the intrusion and interrupts the adversary's actions. These scenarios are used to estimate probability of neutralization.

Step 8. Conduct Probability of Neutralization Analysis. Neutralization analysis provides information about how effective the facility's response force will be at neutralizing the threat using the identified worst-case scenarios. Conducting an actual attack at a facility is generally not possible; therefore, simulations are used to predict the probability of neutralizing violent adversaries after interrupting them. Although it is difficult to accurately model armed conflict between small forces, careful use of appropriate tools and methods can lead to results that indicate the likely performance of the response force if attacked (as defined by the DBT). These tools are used to estimate the

likelihood that a response force will win against an attacking threat before it accomplishes its objective (theft or sabotage).

Tabletop and neutralization analysis, in conjunction with the results of limited scope performance tests conducted at the site, test the knowledge, skills, and abilities of the response forces, and help determine the effectiveness of the response forces against the DBT.

Step 9. Evaluate Physical Protection System Effectiveness and Estimate P_E. The measure of overall PPS effectiveness is the probability of system effectiveness, P_E. System effectiveness is determined quantitatively using the following equation:

$$P_E = P_I * P_N$$

where:

P_I = probability of interruption of the adversary attack by the response force (estimated in Step 7)

P_N = probability of neutralization of the adversary by the response force (estimated in Step 8)

Step 10. Identify Vulnerabilities and Areas for Improvement; Propose Upgrades. The VA team reviews baseline system effectiveness and makes a determination of the effectiveness of the PPS elements in protecting targets against the DBT. If overall system effectiveness meets the determined requirement, then the assessment is complete. Otherwise, system vulnerabilities are identified and areas for improvement are proposed. Both improvements to the PPS functions (detection, delay, and response) and to the overall system are considered for increasing the system effectiveness.

Step 11. Re-evaluate System with Proposed Improvements. Once the upgrades are developed the system is analyzed with the postulated upgrades. The upgrade analysis is conducted and new estimates for probability of interruption, probability of neutralization, and probability of system effectiveness are determined. The process is repeated until upgrades are identified that satisfactorily address system vulnerabilities and increase overall system effectiveness to an acceptable level established by the government.

Step 12. Report Results and Recommendations. The final step of the VA is reporting the results in a manner useable to the decision-makers responsible for PPS decisions. The reporting of results is typically conducted by two methods: briefings and written reports. The goal of the reporting phase is to provide accurate unbiased information that clearly and accurately defines the current system effectiveness of the evaluated PPS and provides potential solutions if the current protection system is deemed ineffective.